

1. Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.

```
Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix  . : bbrouter
    IPv6 Address. . . . . : 2001:db8:1000:1::10
    IPv6 Address. . . . . : 2402:a00:408:5605:b47b:d78:cf2:784d
    Temporary IPv6 Address. . . . . : 2402:a00:408:5605:1c7d:4da4:3d04:bb83
    Link-local IPv6 Address . . . . . : fe80::6962:fd55:6f5d:1781%15
    IPv4 Address. . . . . : 192.168.1.2
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 2001:db8:1000:1::1
                                fe80::6231:92ff:fe04:38bd%15
                                192.168.1.1
```

2. Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.

```
C:\Windows\system32>ping google.com

Pinging google.com [2404:6800:4000:100c::64] with 32 bytes of data:
Reply from 2404:6800:4000:100c::64: time=12ms
Reply from 2404:6800:4000:100c::64: time=12ms
Reply from 2404:6800:4000:100c::64: time=13ms
Reply from 2404:6800:4000:100c::64: time=13ms

Ping statistics for 2404:6800:4000:100c::64:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 12ms, Maximum = 13ms, Average = 12ms
```

3. Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.

```

C:\Windows\system32>tracert flipkart.com

Tracing route to flipkart.com [103.243.32.90]
over a maximum of 30 hops:

  0  1 ms    1 ms    1 ms   RTK_GW.bbrouter [192.168.1.1]
  1  2 ms    2 ms    2 ms   10.230.192.1
  2  3 ms    2 ms    2 ms   10.111.24.5
  3 11 ms   11 ms   11 ms   136.232.112.109
  4  *        *        *      Request timed out.
  5 38 ms   33 ms   33 ms   115.244.220.182
  6  *        *        *      Request timed out.
  7  *        *        *      Request timed out.
  8  *        *        *      Request timed out.
  9  *        *        *      Request timed out.
 10  *        *        *      Request timed out.
 11 33 ms   32 ms   32 ms   103.243.32.90

```

4. Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.

```

C:\Windows\system32>nslookup spotify.com
Server:   UnKnown
Address:  fe80::6231:92ff:fe04:38bd

Non-authoritative answer:
Name:     spotify.com
Addresses: 2600:1901:1:7c5::
           35.186.224.24

```

5. A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check.

Order	Command	What it checks
1	ipconfig	Checks whether the laptop has a valid IP address, subnet mask, default gateway, and DNS server .
2	ping	Tests basic network connectivity . For example, ping 8.8.8.8 checks whether the laptop can reach the Internet.
3	nslookup	Checks whether DNS is resolving Zomato's domain name to an IP address.

Example commands

- ipconfig
- ping 8.8.8.8
- nslookup zomato.com